Thesis Handoff Brief

Investigating the Sim-to-Real Gap in RGB-D Based Tabletop Pose Estimation and Manipulation

Author: Marcell Mor Szokedencsi (s5173191) — BSc Artificial Intelligence, University of Groningen Supervisor: Dr. Hamidreza Kasaei (IRL Lab, Bernoulli Institute, FSE) · Course WBAI901-15 (15 ECTS) · written in Overleaf Document purpose: A complete, self-contained record of the project — research questions, methods, every experiment with exact numbers, the full struggle/solution journey (especially Phase 4C, which exists nowhere else), findings, limitations, the thesis structure, and the reference list (including citations that must not be used). Hand this to a fresh assistant to write the thesis without re-deriving anything.

> How to read this: Parts 0–2 = the thesis as it stood before today (Phases 1–4B). Part 3 = today's work (Phase 4C real manipulation), in full narrative detail. Parts 4–7 = synthesis, the writing plan, references, and where every file lives.

PART 0 — THE THESIS AT A GLANCE

Core contribution (the study): a careful, quantified study of how realistic Time-of-Flight (ToF) depth-sensor noise affects the sim-to-real gap in RGB-D tabletop pose estimation and manipulation. The headline is an honest negative result.

Baseline (explicitly NOT the contribution): the manipulation pick-and-place is the proposal's authorised "predefined grasp strategy for known object types" — a downstream test of whether the pose-null propagates to grasping. All Phase-4C real-manipulation engineering (the closed-loop visual servo, calibration fixes, the USB-HID bridge) is Methods/Appendix material, not a results contribution. The thesis must not present the manipulation system as the novel deliverable.

Research questions (verbatim from `PROJECT_STATUS_REPORT.md` / `SUPERVISOR_MEETING.md`)

- Main RQ: "How does modelling realistic Time-of-Flight (ToF) sensor noise affect the sim-to-real gap in RGB-D tabletop pose estimation and manipulation?"
- SQ1 (noise statistics): "What is the real ToF sensor's noise behaviour, and how well do literature models (Nguyen 2012, Kinect-class) describe it?" → Phase 2
- SQ2 (pose accuracy): "Does injecting measured noise into simulation change pose-estimation accuracy relative to clean simulation?" → Phase 3
- SQ3 (manipulation propagation): "To what extent do pose estimation errors propagate to manipulation performance (pick success rate and placement accuracy)?" → Phases 4A (sim) / 4B (real perception) / 4C (real manipulation)
- Proposal hypothesis (verbatim): "Injecting measured ToF noise into simulation will reduce the sim-to-real gap." — NOT confirmed (this is the negative result; write it as a strength).

The four headline findings

1. Negative result: for a depth-segmented median-centroid estimator at OAK-D-measured noise, injecting measured ToF noise has no measurable effect on pose RMSE ($\Delta < 0.1$ mm vs clean) and none on grasp success. The per-axis median discards the edge flying-pixel outliers that constitute the real noise.
2. Pose threshold ≈ ×10: accuracy only degrades when axial noise is scaled to ≈10× measured ($\sigma \approx 9$ mm) — an order of magnitude above reality.
3. Manipulation collapse ≈ ×20: sim grasp success is flat clean→×10 and collapses only at ×20 (the extra factor exists because grasping is gated by object-vs-gripper tolerance, not perception alone).
4. Real confirms sim: the measured real-vs-sim planar pose gap is −1.4 to +3.6 mm across all four objects (Phase 4B); and the closed-loop real manipulation (Phase 4C, today) gives blue_box 11/15 = 73 % (100 % in the near workspace) vs 100 % sim-feasible — a concrete SQ3 datapoint.

Why the negative result is valid (not a broken pipeline): the injector is verified — scaling the axial term up does eventually break perception (the ×20 collapse). Realistic noise simply sits below the estimator's noise floor.

PART 1 — SYSTEM & METHODS

Hardware

- Arm — Hiwonder xArm 1S (a.k.a. LewanSoul/LeArm), a ~€200 hobby arm. Modelled as a 5-DOF kinematic chain + parallel-jaw gripper: base-yaw (servo 6) + a planar 3R (servos 5/4/3, shared parallel pitch axes) + wrist-roll (servo 2);

gripper fixed, tip +0.09 m. URDF from GrabCAD. ⚠ The status report's file inventory mislabels it "6-DoF"; the IK is 5-DOF — verify against `xarm.urdf` before writing Methods.

- Because the three pitch joints form a planar 3R arm, position is solved in closed-form analytic IK (`arm_ik.py`, pure NumPy, no MoveIt/KDL except optional FK cross-check) → every grasp error is attributable to perception, not solver convergence. Cite Craig, Introduction to Robotics, 3rd ed., ch. 4.
- Geometry quirk: the arm is table-mounted, so a purely vertical top-down grasp is infeasible at most reaches — `feasible_approach` searches the steepest reachable slanted approach within joint limits (this becomes important in Phase 4C: the approach tilt degrades from ~88° near the base to ~70–80° at far reach).
- Real gripper opens to 55 mm (the idealized URDF opened to 72 mm — a disclosed sim-to-real gap source; tightens coke's capture tolerance to 1.5 mm).
- Real control (Phase 4C): USB-HID (VID 0x0483 / PID 0x5750), driven by `pip install xarm` + `hidapi`, servo IDs 1–6 (units 0–1000 ≈ ±125°). Lightweight ROS-free Python bridge on Windows (not ros2_control — USB-HID in WSL needs brittle usbipd). Needs a USB-A↔USB-B cable + DC power on. The real-arm FK was found ~28 % off (URDF link lengths ≠ real) — worked around with a similarity-scale (~0.777) table→arm extrinsic. (This 28 % error is the seed of the entire Phase-4C story.)
- Camera — Luxonis OAK-D SR PoE (ToF): fixed overhead/top-down at 0.68 m (real mount height carried into sim → eliminates eye-in-hand drift, isolating the depth axis). RGB-D 640×480, ~60° FOV. Multi-frequency phase-unwrapping → unusually quiet bulk depth (sub-mm axial σ, flat across range). DepthAI v3.5. Factory intrinsics used (no Zhang calibration run). The PoE link (`169.254.1.222`) is unreachable from WSL's NAT, so it is driven Windows-side.

Software / middleware

- Gazebo Classic 11 + ROS 2 Humble. `ros2_control` (JTC + effort/PID gripper) via a vendored `gazebo_ros2_control`. Arm anchored to `world` by a fixed joint at z = 0.76. Custom MIT `xarm1s_grasp_plugin` (contact-triggered fixed-joint attach, PAL/IFRA pattern) — for visualisation only; measured data come from the analytic grasp-feasible layer. ⚠ Gazebo Classic is end-of-life (Jan 2025) — stable but unsupported.
- Windows/WSL split: authoritative ROS 2 workspace is in WSL2 Ubuntu-22.04 at `/home/szokedencsim/thesis_ws`. For Phase 4C: arm control = Windows `py -3.11` (xarm/hidapi), camera/depthai = Windows `py -3.14`, perception + analysis = WSL `python3`.

Perception pipeline (`pose_estimator.py`) — classical/geometric by design

1. RGB→HSV per-object masks used only to label which blob is which (never for geometry).
2. Depth-based table removal: pixels ≥ 5 mm closer than the median valid depth are "object" (deliberately replaces RANSAC plane fitting — the support plane is fronto-parallel and known).
3. `cv2.connectedComponents` → one blob per object.
4. Per-axis median centroid of the back-projected blob → table-frame position. Translation-only (3-DoF); orientation = identity. The median is what discards the 53–99 % edge flying-pixel population — the crux of the whole thesis.
5. Detections matched to ground truth by greedy nearest-neighbour (a deliberate, documented simplification — NOT Hungarian).
6. ⚠ Code-honesty fix: the `pose_estimator.py` docstring still falsely claims "RANSAC plane segmentation (Open3D)" — must be corrected so code and thesis agree.

The four objects (used in every phase)

Object	Geometry	Width	Height	Jaw capture tol = (55−w)/2	Character
coke_can	slim primitive cylinder	52 mm	47.3 mm	1.5 mm	largest/tallest; fills the jaw
banana	re-centred solid-yellow mesh	35 mm	26.1 mm	10 mm	curved/elongated (12 cm), 120 g
golf_ball	primitive sphere	43 mm	29.6 mm	6 mm	small, round, smooth, 46 g
blue_box	primitive box	40 mm	28.5 mm	7.5 mm	sharp-edged — the only robust object

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PART 2 — PHASES 1–4B (results before today)

Phase 1 — Simulation infrastructure (DONE)

Full ROS 2 Humble / Gazebo Classic 11 stack: xArm URDF, overhead RGB-D camera (640×480, 60° FOV, top-down 0.68 m), tabletop world with the 4 objects, parallel-jaw gripper, perception→evaluator→CSV pipeline. The closed-loop demo was honestly deferred into Phase 4A. Struggles: diagnosing why the sim wouldn't boot took ~a day and uncovered 5 separate bugs all masquerading as "controllers stuck": unset `GAZEBO_MODEL_PATH`; `gzserver` hanging on the online model DB (fixed `GAZEBO_MODEL_DATABASE_URI=''`); a broken local patch to `gazebo_ros2_control`; the arm spawning free-floating and falling off the table (fixed with a `world` fixed joint at z=0.76); a controller-load race (fixed with `OnProcessExit` event chains).

Phase 2 — Real ToF noise characterization (DONE) — SQ1

Sensor: OAK-D SR PoE; 200 frames/capture; raw filters FPPN/optical/wiggle ON, temporal/median OFF (on-camera smoothing disabled to expose raw noise).

(a) Planar protocol (5 distances 0.40–0.80 m, 200 frames, matte fronto-parallel target): axial precision is sub-mm and essentially flat — no Kinect-style range growth:

Distance	0.40	0.50	0.60	0.70	0.80 m
axial σ	0.87	0.80	0.68	0.55	0.69 mm

Bias within ±3 mm. Nguyen's quadratic range model does NOT apply — fitted $\sigma_z(z)=a+b\cdot z^2$ gives a = 0.91 mm, b = −0.000490 m^{-1} (negative). (n = 33 920 px/distance.)

(b) Object protocol — body vs a 6-px edge ring (the real finding): body noise stays quiet; the edge mixed-pixel (flying-pixel) rate is huge on all objects:

Object	body σ	dropout	edge flying-pixel rate
coke_can	0.70 mm	0 %	53.1 % (re-capture 62.5 %, n=2)
banana	0.62 mm	0 %	98.7 %
golf_ball	0.65 mm	0 %	98.2 %
blue_box	0.71 mm	0 %	94.9 %

Findings: the OAK-D's multi-frequency unwrapping gives Kinect-beating bulk depth (0.6–0.7 mm vs 2–10 mm); the pose-relevant noise lives at edges, not the body → motivates the median-centroid estimator + an edge noise term. Caveats: the "sharp→low / curved→high" story is too simple (the sharp box is still 94.9 %); coke is low mainly because it is the largest object; the rate is computed over a fixed 6-px ring → partly size-sensitive (report qualitatively + a ring-width caveat). Coke's gloss adds a small specular dropout (centroid unaffected).

Phase 3 — Noise injection + pose evaluation (DONE) — SQ2

Injector (`depth_noise_injector`) implements four literature terms from measured YAML: axial Gaussian σ_z(z) (Nguyen Eq.3) = 0.9 mm flat (b=0); lateral jitter (Nguyen Eq.1) = 0.8 px (literature value, not separately measured); edge mixed-pixels (Foix/Sarbolandi) = p_edge 0.70; dropout (Sarbolandi) = 0 % measured. Conditions: `sim_clean`, `sim_noisy_oakd`, `sim_noisy_nguyen` + an axial_scale ablation (×1,×2,×5,×10,×20). Re-run 2026-06-07 on the current clean models, ~435 frames/condition, 100 % detection.

Pose RMSE (mm), planar / 3D — from `phase3_pose_table.csv`: | axial_scale | coke | banana | golf | blue_box | |---|---|---|---|---| | 0 (clean) | 3.31 / 4.85 | 8.71 / 10.67 | 4.77 / 7.89 | 3.06 / 3.06 | | ×1 (measured) | 3.45 / 4.99 | 8.60 / 10.64 | 4.66 / 7.94 | 3.44 / 3.44 | | ×5 | 3.53 / 6.03 | 8.68 / 11.12 | 4.53 / 8.55 | 3.63 / 3.71 | | ×10 | 3.47 / 7.41 | 9.08 / 12.19 | 4.39 / 10.48 | 3.44 / 4.03 | | ×20 | 9.55 / 18.43 | 22.43 / 24.53 | 8.58 / 19.08 | 8.68 / 10.21 |

Findings: (1) measured (×1) ≈ clean ≈ nguyen — no effect through ×10; planar XY stays flat ~3.5 mm; the small 3D creep is the axial/Z component. (2) Robustness threshold ≈ ×10 (σ ≈ 9 mm), all four collapse together at ×20. The old "noise regularises coke" claim was an artefact of the off-centre textured mesh — gone with the clean slim can (coke 80→~5 mm, banana 39→~11 mm 3D RMSE).

Phase 4A — Simulated pick-and-place (DONE) — SQ3 (sim)

Metric = analytic `grasp_feasible`: a grasp succeeds iff the perceived centre is within `capture_tol` of the true centre (open jaws would straddle it). No physical contact scored — Gazebo Classic cannot reliably contact-grasp small/curved objects (proven: finger collisions pass through; object pose bit-identical before/after closing). Renamed `pick_success`→`grasp_feasible` to avoid overclaiming. Gripper corrected 72→55 mm (real measured). Experiment — varied-placement sweep, 480 analytic trials: each object alone at 20 seeded positions+orientations over the reachable annulus (r 0.12–0.30 m, full yaw, IK-validated; coke lying), × 6 noise conditions = 6×4×20.

Success rate (%) — from `phase4a_success_table.csv` (N=20/cell): | Condition | coke (1.5mm) | banana (10mm) | golf (6mm) | blue_box (7.5mm) | overall | |---|---|---|---|---|---| | clean | 0 % | 65 % | 65 % | 100 % | 57.5 % | | ×1 measured | 5 % | 65 % | 65 % | 100 % | 58.75 % | | ×10 | 0 % | 65 % | 75 % | 100 % | 60 % | | ×20 | 5 % | 30 % | 20 % | 50 % | 26.25 % |

Findings: all 4 detected 20/20 (accuracy, not detection, is the bottleneck); the 3 graspable objects flat clean→×10, collapse at ×20 — the manipulation echo of the pose threshold. Per-object governed by raw clean error vs jaw tolerance: blue_box 3.5 vs 7.5 → 100 %; golf 5.2 vs 6.0 → 65 % (on the boundary); banana 9.3 vs 10 → 65 % (on the boundary); coke 4.9 vs 1.5 → 0 % (floored — at the gripper size limit, regardless of noise). Struggles: the banana was 0 % at every level — a model artefact, not noise: (1) Gazebo ignored the SDF material on the textured mesh → HSV labeller failed → swapped to a solid-yellow mesh; (2) the visual mesh was ~28 mm off-centre from its link origin (an offset that rotates with yaw, so no fixed bias fixes it) → fixed with a link-frame `<pose>` = −C_mean → error 28.7→10.6 mm, rate 0→65 %. Plus the documented Gazebo grasp-realism work (contact-triggered weld, effort fingers, fingertip pads, physics tuning) and an intermittent GUI-only banana grab-fling (an ODE solver-timing effect at joint formation; not reproducible headless; cosmetic — data is analytic).

Phase 4B — Real-world perception / sim-to-real pose gap (DONE) — SQ3 (real perception)

Setup (offline route): fiducials detected on the OAK-D RGB camera (CAM_B), not the ToF amplitude (the soft amplitude gave 0/24 corners) → composed into the depth frame via the factory CAM_A↔CAM_B extrinsic. Camera→table via a printed 7×5 ChArUco board (DICT_5X5_100, 100 mm squares): direct `solvePnP` (no hand-eye cal), 20/24 corners @ 0.26 px reproj; height 0.683 m vs depth-plane fit 0.684 m (1.4 mm agreement). 5 real scenes, same perception pipeline, matched to ArUco truth. Metric = planar (table-XY) error (flat marker gives XY truth, no top-Z).

Planar pose RMSE (mm) — from `phase4b_gap_table.csv`: | Condition | coke | banana | golf | blue_box | |---|---|---|---|---| | sim_clean (n=20) | 10.44 | 13.27 | 6.15 | 3.78 | | sim_noisy_oakd (n=20) | 10.35 | 13.24 | 6.02 | 3.54 | | real (n=4) | 9.38 | 12.13 | 4.75 | 7.42 | | GAP (real − sim_clean) | −1.1 | −1.1 | −1.4 | +3.6 |

Findings: the real-vs-sim planar pose gap is just −1.4 to +3.6 mm across all four objects — sim ≈ real; `sim_clean ≈ sim_noisy_oakd` still holds → the negative result carries end-to-end on real data. Limitation: real is n=4/object — in 4 of 5 scenes one object drifted into the ToF edge zone and failed to segment (the ToF reliable area is smaller than the RGB marker FOV).

PART 3 — PHASE 4C: REAL MANIPULATION (today's work, in full)

> This is the part documented nowhere else. It answers SQ3 on the real arm (does the pose-null propagate to a real grasp rate?) and the engineering is baseline/Methods/Appendix, not the contribution. The story matters because the struggle (and how it was overcome) is itself a documented sim-to-real finding.

Starting point

The committed real executor (`real_grasp_executor.py`, HEAD `6e592ce`) used the perceived PCA grasp axis. To make the real and sim grasp recipes identical (so the measured gap is pure execution, not a perception confound), it was switched to always use the predetermined per-object nominal grasp angle (same as sim) — a one-line change (`minor_table = d.get("minor_axis")`) plus a re-added constant `--off-x/--off-y` aim-nudge flag. The sim path (`grasp_executor.py`, `grasp_common`, `arm_ik`) was left byte-identical.

The problem that triggered everything: a ~4 cm working zone

On the bench, the 40 mm `blue_box` (needs the grasp centred within ±7.5 mm) grasped reliably only in a ~4 cm × 4 cm patch of the table — despite the arm's ~20 cm reach. The data (`real_manip_results.csv`) showed: detection ✓, IK feasible ✓, jaws close ✓ — but the jaws closed to ~17 mm on empty air (the box is 40 mm) → they were missing the box body, only ~3/15 holding.

The misdiagnosis trail (documented because it's instructive)

1. Not the grasp angle — a wrong jaw angle would stall the jaws near 55 mm (can't close across the 60 mm length), not at 17 mm. 17 mm = empty jaws = a positional miss.
2. The `--off-x` nudge was correcting the WRONG axis. The table→arm calibration rotation `R_arm_from_table` ≈ 85° (`[[0.094,−0.996],[0.996,0.094]]`), so an arm-frame `--off-x` moved the aim along the table-Y axis (the box's closing axis, ±7.5 mm tolerance) and did nothing for the table-X miss the operator saw — `--off-x −10` was actually a ~13 mm table-Y perturbation hurting the grip. Fix: apply the offset in the table frame (before `table_to_arm`). Verified by dry-run: +50 mm table-X → +38 mm arm-Y, matching `scale·R·[0.05,0]`.
3. Then: a position-dependent error a constant offset cannot fix. With table-X corrected, the operator reported the Y miss flipped sign across the table (left tip hits on the left side, right tip on the right). A single offset can't push one way on the left and the other on the right → the error is position-dependent.

The honest dead-end, and the user's correct push-back

At this point the assessment was "this is the open-loop accuracy floor of a cheap arm; measure it honestly." The student rightly pushed back: a ~4 cm zone on an arm with ~20 cm reach is not plausible for a thesis, and a reference video proves the hardware can be precise. This was the turning point — the 4 cm zone was an artefact of our own shortcut, not a hardware limit.

The multi-agent investigation (19 agents) — root cause, verified

A research+redesign workflow (web research across GitHub/Scholar/StackOverflow + redesign + adversarial critique + a project-manager synthesis) verified the root cause directly against the live data: - The error is an unidentifiable, position-dependent kinematic-model error. The real arm over-reaches the URDF/`arm_ik` model by ~27 %. The global-scale workaround (`arm = scale·R@table + t`) leaves 14.1 mm RMS / 24.9 mm max on all 6 touch points (the published 7.94 mm only survives by dropping a point to n=5). 14.1 mm > the box's ±7.5 mm tolerance → only a narrow band works. - 6 touch points cannot identify a 13-parameter kinematic model (overfits, no holdout). The "fit the per-servo gain first" hypothesis was falsified (gain-only fits leave 27–37 mm, worse). - The approach-tilt limit is edge-only (vertical ~88° holds to r≈0.19 m; tilts to 85→75→65° past r≈0.20 m) — not the dominant cause. - ⇒ Any open-loop calibration is capped at ~8–14 mm and cannot clear ±7.5 mm workspace-wide. So: close the loop.

The solution: closed-loop visual nudge (FK-independent)

Tape one ArUco marker (DICT_4X4_50, id 4 — the object markers use ids 0–3) flat on the gripper so the overhead OAK-D sees it. Before descending, iterate, in the table frame:

```
err = obj_xy − (gripper_marker_xy + tip_offset)
target += K · err          (K ≈ 0.6)
```

until `|err| < 3 mm`, then descend + close. Because `err` is a difference of two points measured through the SAME camera in the SAME frame, the wrong FK and the 0.777 scale cancel as common-mode bias — they only need to be good enough to converge. Grasp accuracy becomes camera-bound (~1–4 mm), workspace-wide, independent of the broken kinematic model. This is the same idea as a chess-playing arm that verifies against known points rather than trusting a perfect model — and it ties straight into the sim-to-real theme (open-loop calibration error is the dominant real-world gap; one visual-feedback correction closes it within its working zone). (A dense camera-measured thin-plate-spline correction field — REPLAB, arXiv:1905.07447 — was designed as a citable open-loop coarse stage + fallback, but was deferred: the closed loop alone delivered the result.)

Implementation (all UNCOMMITTED; `arm_ik` constants deliberately untouched because they are shared with the committed sim Phases 2–4A)

- `xarm1s_manipulation/xarm1s_manipulation/visual_servo.py` — the pure, hardware-free convergence core `servo_align(obj_xy, observe, command, …)`. Returns `{converged, measured, misses, final_target, history}`. An offline self-test with a synthetic position-dependent bias proves it: 500/500 random objects converge 9.2 mm → 1.85 mm (≤4 mm 100 %) in ~2.7 iterations.
- `tof_perception/scripts/detect_gripper_marker.py` — RGB→table-frame gripper-marker detector, reusing the proven `ingest_real_scene` path (`detect_markers → marker_pose(solvePnP) → T_table_rgb @ pose`). Validated on a real scene (4 object markers → correct table-XY) and on the live gripper marker (`found:true`).
- `real_grasp_executor.py --servo` — wires the loop into `run_one` between the pre-grasp hover and the descent; additive (open-loop path byte-identical with `--servo` off). Flags: `--servo`, `--servo-id 4`, `--servo-len 0.05`, `--servo-gain 0.6`, `--servo-thresh 3`, `--servo-iters`, `--tip-dx/--tip-dy`. The bench observer captures an RGB (Win py-3.14 subprocess) → detects (WSL python3 subprocess), with a freshness guard (delete-before-capture + mtime/returncode check) and capture retries (the OAK-D drops X_LINK on rapid re-opens). A synthetic observer drives the full path under `--dry-run`.

Adversarial code review (10 agents) — 3 real bench bugs found and fixed

1. Detection-miss faked convergence — a missed marker returned `err=0`, which the loop read as "converged" and descended at the uncorrected aim. Fixed: a miss now returns `None` → the loop holds aim, retries, and never counts a miss as convergence.
2. Stale `_servo_rgb.png` re-detected as fresh — a failed capture re-used the previous image. Fixed: delete-before-capture + mtime/returncode freshness guard + capture retry-with-delay.
3. `--off-x/--off-y` corrupted the logged provenance under `--servo` (didn't affect the grasp). Fixed: gated on `servo is None`. (A 4th finding — the predetermined-angle change is not byte-identical to HEAD — was the intended sim-parity change, not a bug.)

The bench journey to a clean result

- First real `--servo` grasps at the tilted far edge (tilt 77–82°) diverged (the high/offset marker swings as the arm reconfigures).
- A camera X_LINK drop on a later step exposed the miss-convergence bug (now fixed).

- In the vertical zone (tilt ~88°), with `--servo-iters 8`, it converges and grasps. Example: `err 19.1 → 0.96 mm conv`, gripped even with the box deliberately rotated.

The result — Phase-4C blue_box formal run (5 placements × 3 = 15 trials)

`blue_box 11/15 = 73 % lift-and-hold (Wilson 95 % CI 48–89 %).` Clean labeled data: `evaluation/results/real_manip_4c_formal.csv`; Results 5.5 artifacts: `evaluation/results/phase4c_real_table.{txt,csv}` + `phase4c_real_vs_pred.png`.

placement	reach r	servo err (mm)	3 reps
p1 (0.35,0.16)	0.20 m	20→2, 23→2, 23→2	[OK][OK][OK]
p2 (0.31,0.22)	0.22 m	26→7, 31→8, 24→2	[OK][X][OK]
p3 (0.37,0.29)	0.16 m	16→2, 17→2, 16→3	[OK][OK][OK]
p4 (0.30,0.34)	0.23 m	44→17, 40→11, 40→11	[X][X][X]
p5 (0.37,0.37)	0.19 m	34→2, 31→2, 33→3	[OK][OK][OK]

- Every failure is at r ≥ 0.22 m. Within the validated working zone r ≤ 0.20 m (p1, p3, p5): 9/9 = 100 %.
- Sim-to-real (the SQ3 datapoint): sim analytic feasibility 100 % → real closed-loop 73 % (100 % in-zone). The Phase-4B predictor (planar err 7.4 mm vs jaw tol 7.5 mm = "marginal") called it: open-loop would land on the cliff (≈0 % across a spread, the old 4 cm zone); the closed loop pushed it to 73 %/100 %-in-zone.
- The before/after IS the Results 5.5 figure: open-loop ~4 cm patch → closed-loop near-workspace.

Limitations + the known fix

- Far-reach falloff (r ≳ 0.20 m): the loop gain collapses because the marker is mounted high and behind the jaws — at extended reach it swings instead of tracking the fingertips. The fix (deferred): mount the marker low and directly over the jaw gap (so it tracks the fingertip ~1:1 → converges full-reach). The near-zone result already proves the method.
- Camera flakiness: the OAK-D drops X_LINK / throws crash-dumps on rapid re-opens; the retry + freshness guard absorb it, but a persistent-camera architecture would be cleaner.
- Other objects (predicted, not yet run): banana (elongated like the box, ±10 mm tol — more forgiving) should work closed-loop in the zone — nearly free; golf (4.8 < 6 mm; closed-loop fixes the aim that was its wall, round-slip the only residual risk) worth a try; coke = a SIZE WALL (sim-feasible 0 %, 52 mm body in a 55 mm jaw) — closed-loop cannot fix it; document, don't chase.
- Data-hygiene caveats: the formal trials in the main `real_manip_results.csv` are mislabeled `placement_id=1, repeat=1` (the run template's flags weren't bumped), and `summarize_phase4c.read_real` counts every row with no placement filter → always summarize the clean file `real_manip_4c_formal.csv` (`summarize_phase4c.py --csv real_manip_4c_formal.csv`).

PART 4 — OVERALL FINDINGS & NARRATIVE

1. The negative result (headline): measured ToF noise has no measurable effect on pose RMSE (Δ<0.1 mm) or grasp success — the per-axis median discards the 53–99 % edge flying-pixel outliers. Refutes the proposal hypothesis.
2. Pose threshold ≈ ×10 (σ≈9 mm); manipulation collapse ≈ ×20; both far above reality. The extra factor before manipulation breaks exists because grasp is gated by object-vs-gripper tolerance.
3. The injector is verified (×20 collapse) → the null is sound, not a broken pipeline.
4. Per-object spectrum (size vs shape): blue_box robust (margin to spare); golf/banana borderline ~65 % (error sits on the tolerance); coke floored 0 % — a size limit (52 mm in 55 mm jaw), not curvature.
5. The phases connect: Phase 2 reframed the problem (real ToF is quiet in the bulk, noisy at edges) → Phase 3 (negative result) → Phase 4A (same shape in sim manipulation) → Phase 4B (sim≈real on real data, −1.4..+3.6 mm) → Phase 4C (real manipulation rate via closed-loop; the open-loop calibration error, not depth noise, is the real-world bottleneck). Synthesis: for a robust median-centroid estimator the sim-to-real gap is NOT governed by depth-noise-model fidelity — it lives in calibration, lighting, and segmentation.

Limitations / open issues (thesis-level)

- Gazebo Classic cannot reliably contact-grasp small/curved objects → the sim success metric is analytic/geometric (a key supervisor discussion item).
- Noise model omits spatial correlation and multipath; lateral jitter uses the Nguyen literature value (0.8 px), not separately measured.
- Phase-4A CIs are Wald (collapse to ±0 at 0 %/100 %) — Wilson would be more honest (Phase 4C already uses Wilson).
- Real Phase-4B is n=4/object (ToF reliable zone < RGB marker FOV).

- Phase-4C real manipulation is currently blue_box only, in a near-zone (r≤0.20 m); the far-reach limit is a documented marker-mount issue.
- Gazebo Classic is end-of-life (Jan 2025).

Future work

Run banana + golf with the closed loop; remount the marker low for full-reach convergence; re-solve the arm-base calibration with more points / a learned correction field; (out of scope) model spatial correlation + multipath; close the loop fully (visual servoing) as the principled accuracy ceiling.

PART 5 — THESIS WRITING GUIDE

Template: official RUG BSc-AI Overleaf template (`bachelorproject.sty`, two-column, FSE-branded). Written in Overleaf (not in this repo). Citations IEEE-numbered (`\bibliographystyle{IEEEtran}` — the AI template defaults to APA/`apacite`; confirm with Kasaei). Add a declaration of originality / AI-use. Two open confirmations with the supervisor: (1) AI vs CS program/template; (2) APA vs IEEE.

Ch	Title	Contents	Writable now?
1	Introduction	reality gap, RQ + 3 SQs, hypothesis, contributions, outline	[OK]
2	Related Work	sim-to-real/domain randomization; ToF/depth noise; tabletop perception; 6D pose eval; grasping (cite MVGrasp)	[OK]
3	System & Methods	sim/URDF/overhead RGB-D; noise-char protocol; injection model; depth-seg perception (median, translation-only); analytic IK; Phase-4B ChArUco extrinsic; Phase-4C closed-loop servo (baseline/appendix)	[OK]
4	Experimental Setup	hardware, 4 objects, conditions, trial counts, metrics (translation RMSE/MAE, grasp-feasible, Wilson)	[OK]
5	Results	5.1 noise char · 5.2 injection (Δ<0.1 mm) · 5.3 ablation (≈×10) · 5.4 sim manipulation · 5.5 real pose gap (4B) + real manipulation (4C: 73 %/100 %-in-zone)	5.1–5.4 [OK]; 5.5 mostly in
6	Discussion	negative result valid; median filter eats realistic noise; gap = calibration/lighting; open-loop FK error is the real-world bottleneck (Phase-4C); closing the loop fixes it; Gazebo grasp limitation	draft now
7	Conclusion & Future Work	—	after remaining 4C

Recommended writing order: Ch.3 → Ch.4 → Ch.5.1–5.4 → Ch.2 → Ch.1 → Ch.6 draft → Ch.5.5 + Ch.7. Plug numbers straight from: `phase3_pose_table.{txt,csv}`, `phase4a_success_table.csv`, `phase4b_gap_table.csv`, and `summarize_phase4c.py --csv real_manip_4c_formal.csv`. Figures already exist (σ-vs-z, edge overlays, ablation curve, `phase3_pose_vs_noise.png`, `phase4c_real_vs_pred.png`).

PART 6 — REFERENCES (`thesis_writing/references.bib`)

Used by the code/experiments (PART 1 — safe to cite): - `nguyen2012modeling` (axial+lateral injector eqs; the range model Phase 2 refutes); `khoshelham2012accuracy` (planar protocol); `sarbolandi2015kinect` (edge mixed-pixels + dropout); `foix2011lockin` (lock-in ToF / flying pixels). - `tobin2017domain` (domain randomization / seeded RNG ablations); `hodan2018bop` (BOP — cite ONLY for the visible-top-surface alignment, NOT ADD/VSD). - `garridojurado2014aruco`, `romeroramirez2018charuco` (ArUco/ChArUco truth + extrinsic); `foote2013tf` (tf2); `craig2005robotics` (closed-form IK). - `koenig2004gazebo`, `macenski2022ros2`, `bradski2000opencv` (Gazebo / ROS 2 / OpenCV).

Related-work framing (PART 2): `jakobi1995noise` (origin of "reality gap"); `kasaei2021mvgrasp` (supervisor's MVGrasp); `bohg2014grasp` (grasp survey — positions the predefined grasp, not implemented); `rusu2008towards` (classic RANSAC-plane pipeline — cite as contrast only); `pinto2024flexx2` (PMD Flexx2 ToF — corroboration). If Methods calls the median "outlier-robust," add Huber, Robust Statistics (1981) (median 50 % breakdown).

Software/assets (PART 3): `gazebo_ros2_control`, `link_attacher` (PAL/IFRA), `luxonis_oakd`, `depthai_docs`. TODO (only the user can recover): `grabcad_xarm`, `sketchfab_cokecan`, `sketchfab_banana` — verify CC-BY licenses; see `src/ATTRIBUTION.md`.

[DO NOT CITE] DO NOT CITE (PART 4 — would be false claims about the implementation): `kuhn1955hungarian` (eval is greedy NN), `tsai1989handeye` (direct solvePnP, no hand-eye), `zhang2000calibration` (factory intrinsics), `hinterstoisser2012add` (translation-only pose), `fischler1981ransac` + `zhou2018open3d` (median-depth threshold, not RANSAC/Open3D), `olson2011apriltag` (ArUco chosen), `kumra2020antipodal` (learned grasping not implemented), `belhedi2015` (drop).

PART 7 — DATA, FILES & HONESTY CHECKLIST

Results tables/figures (`src/evaluation/results/`): `phase3_pose_table.{csv,txt}`, `phase4a_success_table.csv`, `phase4b_gap_table.{csv,txt}`, `real_manip_4c_formal.csv` (clean Phase-4C), `phase4c_real_table.{csv,txt}` + `phase4c_real_vs_pred.png`, `phase3_pose_vs_noise.png`. Phase-2 params: `src/tof_perception/scripts/fit_results_truly_raw/noise_params.json`; per-object `object_results/object_noise.json`. Notes: `src/xarm1s_manipulation/{PHASE4A_NOTES.md, PHASE4A_RERUN_NOTES.md, GRASP_REALISM_NOTES.md, GRASP_RESEARCH.md, PHASE4C_NOTES.md, PHASE4C_RUNBOOK.md, PHASE4C_VISUAL_SERVO_PLAN.md}`; the 25-pp `thesis_ws/PROJECT_STATUS_REPORT.{md,pdf}`. Core code: `tof_perception/{pose_estimator.py, scripts/ingest_real_scene.py, scripts/detect_gripper_marker.py}`, `noise_injection/depth_noise_injector.py`, `evaluation/{pose_evaluator.py, scripts/summarize_phase4c.py}`, `xarm1s_manipulation/xarm1s_manipulation/{grasp_executor.py, grasp_common.py, arm_ik.py, visual_servo.py, real_arm_driver.py}`, `xarm1s_manipulation/scripts/real_grasp_executor.py`, `xarm1s_grasp_plugin/src/link_attacher.cpp`.

Code-honesty fixes before submission: 1. `pose_estimator.py` docstring falsely says "RANSAC plane segmentation (Open3D)" — correct it (it's a median-depth threshold). 2. Verify the arm DOF (5-DOF IK vs the report's "6-DoF" file label) against `xarm.urdf`. 3. Fill the 3 TODO asset citations in `references.bib` (GrabCAD arm, Sketchfab coke/banana) — see `ATTRIBUTION.md`. 4. The Phase-4C closed-loop code is uncommitted (the student handles all git).

End of brief. Everything needed to write the thesis is above; the Phase-4C narrative and the don't-cite list are the two things you won't find in the older `PROJECT_STATUS_REPORT.md`.